

ASSIGNMENT 1

1. WAQTD display all the details from EMP table

```
SELECT *  
  
FROM emp;
```

2. WAQTD names of all the employees

```
SELECT ename  
  
FROM emp;
```

3. WAQTD name and salary of all the employees

```
SELECT ename, sal  
  
FROM emp;
```

4. WAQTD name and commission of all the employees

```
SELECT ename, comm  
  
FROM emp;
```

5. WAQTD employee id and department number of all the employees

```
SELECT empno, deptno  
  
FROM emp;
```

6. WAQTD name and hiredate of all the employees

```
SELECT ename, hiredate  
  
FROM emp;
```

7. WAQTD name and designation of all the employees

```
SELECT ename, job  
  
FROM emp;
```

8. WAQTD name, job and salary of all the employees

```
SELECT ename, job, sal  
  
FROM emp;
```

9. WAQTD department names present in DEPT table

```
SELECT dname  
  
FROM dept;
```

10. WAQTD department name and location present in DEPT table

```
SELECT dname, loc  
FROM dept;
```

EXPRESSIONS & ALIAS

11. WAQTD name of the employee along with their annual salary

```
SELECT ename, sal*12 AS annual_salary  
FROM emp;
```

12. WAQTD ename and job with their half-term salary

```
SELECT ename, job, sal/2 AS half_term_salary  
FROM emp;
```

13. WAQTD all the details of employees with an annual bonus of 2000

```
SELECT emp.*, sal*12 + 2000 AS annual_bonus_salary  
FROM emp;
```

14. WAQTD name, salary and salary with a hike of 10%

```
SELECT ename, sal, sal*1.10 AS salary_with_hike  
FROM emp;
```

15. WAQTD name and salary with deduction of 25%

```
SELECT ename, sal, sal*0.75 AS salary_after_deduction  
FROM emp;
```

16. WAQTD name and salary with monthly hike of 50

```
SELECT ename, sal, sal+50 AS salary_after_hike  
FROM emp;
```

17. WAQTD name and annual salary with deduction of 10%

```
SELECT ename, sal*12 AS annual_salary,  
sal*12*0.90 AS annual_salary_after_deduction  
FROM emp;
```

18. WAQTD total salary given to each employee (SAL + COMM)

```
SELECT ename, sal + NVL(comm,0) AS total_salary
```

FROM emp;

19. WAQTD details of all the employees along with annual salary

SELECT emp.*, sal*12 AS annual_salary

FROM emp;

20. WAQTD name and designation along with 100 penalty in salary

SELECT ename, job, sal-100 AS salary_after_penalty

FROM emp;

ASSIGNMENT 2

1. WAQTD the annual salary of the employee whose name is SMITH

SELECT sal*12 AS annual_salary

FROM emp

WHERE ename = 'SMITH';

2. WAQTD name of the employees working as CLERK

SELECT ename

FROM emp

WHERE job = 'CLERK';

3. WAQTD salary of the employees who are working as SALESMAN

SELECT sal

FROM emp

WHERE job = 'SALESMAN';

4. WAQTD details of the employee who earns more than 2000

```
SELECT *  
FROM emp  
WHERE sal > 2000;
```

5. WAQTD details of the employee whose name is JONES

```
SELECT *  
FROM emp  
WHERE ename = 'JONES';
```

6. WAQTD details of the employee who was hired after 01-JAN-81

```
SELECT *  
FROM emp  
WHERE hiredate > '01-JAN-81';
```

7. WAQTD name and salary along with annual salary if annual salary is more than 12000

```
SELECT ename, sal, sal*12 AS annual_salary  
FROM emp  
WHERE sal*12 > 12000;
```

8. WAQTD empno of the employees working in department 30

```
SELECT empno  
FROM emp  
WHERE deptno = 30;
```

9. WAQTD ename and hiredate of employees hired before 1981

```
SELECT ename, hiredate  
FROM emp  
WHERE hiredate < '01-JAN-81';
```

10. WAQTD details of the employees working as MANAGER

```
SELECT *  
FROM emp  
WHERE job = 'MANAGER';
```

11. WAQTD name and salary of employees earning commission of 1400

```
SELECT ename, sal  
FROM emp  
WHERE comm = 1400;
```

12. WAQTD details of employees having commission more than salary

```
SELECT *  
FROM emp  
WHERE comm > sal;
```

13. WAQTD empno of employees hired before the year 1987

```
SELECT empno  
FROM emp  
WHERE hiredate < '01-JAN-87';
```

14. WAQTD details of employees working as ANALYST

```
SELECT *  
FROM emp  
WHERE job = 'ANALYST';
```

15. WAQTD details of employees earning more than 2000 rupees per month

```
SELECT *  
FROM emp
```

WHERE sal > 2000;

ASSIGNMENT 3

1. WAQID details of the employees working as CLERK and earning less than 1500

```
SELECT *  
FROM emp  
WHERE job = 'CLERK'  
AND sal < 1500;
```

2. WAQID name and hiredate of the employees working as MANAGER in dept 30

```
SELECT ename, hiredate  
FROM emp  
WHERE job = 'MANAGER'  
AND deptno = 30;
```

3. WAQID details of the employees along with annual salary

(Dept 30, working as SALESMAN, annual salary > 14000)

```
SELECT emp.*, sal*12 AS annual_salary  
FROM emp  
WHERE deptno = 30  
AND job = 'SALESMAN'  
AND sal*12 > 14000;
```

4. WAQID all the details of the employees working in dept 30 or as ANALYST

```
SELECT *  
FROM emp
```

WHERE deptno = 30

OR job = 'ANALYST';

5. WAQID names of the employees whose salary is less than 1100 and designation is CLERK

SELECT ename

FROM emp

WHERE sal < 1100

AND job = 'CLERK';

6. WAQID name, salary, annual salary and deptno

(Dept 20, earning > 1100, annual salary > 12000)

SELECT ename, sal, sal*12 AS annual_salary, deptno

FROM emp

WHERE deptno = 20

AND sal > 1100

AND sal*12 > 12000;

7. WAQID empno and names of the employees working as MANAGER in dept 20

SELECT empno, ename

FROM emp

WHERE job = 'MANAGER'

AND deptno = 20;

8. WAQID details of employees working in dept 20 or 30

SELECT *

FROM emp

WHERE deptno IN (20, 30);

9. WAQID details of employees working as ANALYST in dept 10

```
SELECT *  
FROM emp  
WHERE job = 'ANALYST'  
AND deptno = 10;
```

10. WAQID details of employee working as PRESIDENT with salary of 4000

```
SELECT *  
FROM emp  
WHERE job = 'PRESIDENT'  
AND sal = 4000;
```

ASSIGNMENT 4

1. WAQID names, empno and job of employees working as CLERK in dept 10 or 20

```
SELECT ename, empno, job  
FROM emp  
WHERE job = 'CLERK'  
AND deptno IN (10, 20);
```

2. WAQID names of employees working as CLERK or MANAGER in dept 10

```
SELECT ename  
FROM emp  
WHERE job IN ('CLERK', 'MANAGER')  
AND deptno = 10;
```

3. WAQID names of employees working in dept 10, 20, 30, 40

```
SELECT ename  
FROM emp
```


WHERE deptno IN (10, 20, 30, 40);

4. WAQID details of employees with empno 7902, 7839

```
SELECT *  
FROM emp  
WHERE empno IN (7902, 7839);
```

5. WAQID details of employees working as MANAGER or SALESMAN or CLERK

```
SELECT *  
FROM emp  
WHERE job IN ('MANAGER', 'SALESMAN', 'CLERK');
```

6. WAQID names of employees hired after 1981 and before 1987

```
SELECT ename  
FROM emp  
WHERE hiredate BETWEEN '01-JAN-81' AND '31-DEC-86';
```

7. WAQID details of employees earning more than 1250 but less than 3000

```
SELECT *  
FROM emp  
WHERE sal BETWEEN 1251 AND 2999;
```

8. WAQID names of employees hired after 1981 into dept 10 or 30

```
SELECT ename  
FROM emp  
WHERE hiredate > '31-DEC-81'  
AND deptno IN (10, 30);
```

9. WAQID names along with annual salary of employees

(Working as MANAGER or CLERK in dept 10 or 30)

```
SELECT ename, sal*12 AS annual_salary  
FROM emp  
WHERE job IN ('MANAGER', 'CLERK')  
AND deptno IN (10, 30);
```

10. WAQID all details along with annual salary

(Salary between 1000 and 2000 OR annual salary > 15000)

```
SELECT emp.*, sal*12 AS annual_salary  
FROM emp  
WHERE sal BETWEEN 1000 AND 2000  
OR sal*12 > 15000;
```

ASSIGNMENT 5

1. WAQTD all the employees whose commission is NULL

```
SELECT *  
FROM emp  
WHERE comm IS NULL;
```

2. WAQTD all the employees who don't have a reporting manager

```
SELECT *  
FROM emp  
WHERE mgr IS NULL;
```

3. WAQTD all the SALESMEN in department 30

```
SELECT *  
FROM emp  
WHERE job = 'SALESMAN'
```

AND deptno = 30;

4. WAQTD all the SALESMEN in dept 30 having salary greater than 1500

```
SELECT *  
FROM emp  
WHERE job = 'SALESMAN'  
      AND deptno = 30  
      AND sal > 1500;
```

5. WAQTD all the employees whose name starts with 'S' or 'A'

```
SELECT *  
FROM emp  
WHERE ename LIKE 'S%'  
      OR ename LIKE 'A%';
```

6. WAQTD all the employees except those working in dept 10 & 20

```
SELECT *  
FROM emp  
WHERE deptno NOT IN (10, 20);
```

7. WAQTD employees whose name does NOT start with 'S'

```
SELECT *  
FROM emp  
WHERE ename NOT LIKE 'S%';
```

8. WAQTD all the employees who are having reporting managers in dept 10

```
SELECT *  
FROM emp  
WHERE mgr IS NOT NULL
```

AND deptno = 10;

9. WAQTD all the employees whose commission is NULL and working as CLERK

```
SELECT *  
FROM emp  
WHERE comm IS NULL  
AND job = 'CLERK';
```

10. WAQTD all the employees who don't have a reporting manager in dept 10 or 30

```
SELECT *  
FROM emp  
WHERE mgr IS NULL  
AND deptno IN (10, 30);
```

ASSIGNMENT 6

1. WAQTD number of employees working in each department except PRESIDENT

```
SELECT deptno, COUNT(*) AS no_of_employees  
FROM emp  
WHERE job <> 'PRESIDENT'  
GROUP BY deptno;
```

2. WAQTD total salary needed to pay all the employees in each job

```
SELECT job, SUM(sal) AS total_salary  
FROM emp  
GROUP BY job;
```

3. WAQTD number of employees working as MANAGER in each department

```
SELECT deptno, COUNT(*) AS no_of_managers
FROM emp
WHERE job = 'MANAGER'
GROUP BY deptno;
```

**4. WAQTD average salary needed to pay employees in each department
(excluding employees of deptno 20)**

```
SELECT deptno, AVG(sal) AS avg_salary
FROM emp
WHERE deptno <> 20
GROUP BY deptno;
```

5. WAQTD number of employees having character 'A' in their names in each job

```
SELECT job, COUNT(*) AS no_of_employees
FROM emp
WHERE ename LIKE '%A%'
GROUP BY job;
```

6. WAQTD number of employees and average salary

(salary > 2000 in each department)

```
SELECT deptno,
      COUNT(*) AS no_of_employees,
      AVG(sal) AS avg_salary
FROM emp
WHERE sal > 2000
GROUP BY deptno;
```

7. WAQTD total salary needed to pay and number of SALESMEN in each department

```
SELECT deptno,  
       COUNT(*) AS no_of_salesmen,  
       SUM(sal) AS total_salary  
FROM emp  
WHERE job = 'SALESMAN'  
GROUP BY deptno;
```

8. WAQTD number of employees and maximum salary in each job

```
SELECT job,  
       COUNT(*) AS no_of_employees,  
       MAX(sal) AS max_salary  
FROM emp  
GROUP BY job;
```

9. WAQTD maximum salary given to an employee working in each department

```
SELECT deptno, MAX(sal) AS max_salary  
FROM emp  
GROUP BY deptno;
```

10. WAQTD number of times the salaries are present in EMP table

```
SELECT sal, COUNT(*) AS no_of_times  
FROM emp  
GROUP BY sal;
```

ASSIGNMENT 7

1. WAQTD deptno and number of employees (if there are at least 2 CLERKS in each dept)

```
SELECT deptno, COUNT(*) AS no_of_employees
FROM emp
WHERE job = 'CLERK'
GROUP BY deptno
HAVING COUNT(*) >= 2;
```

2. WAQTD deptno and total salary needed to pay all employees (if there are at least 4 employees in each dept)

```
SELECT deptno, SUM(sal) AS total_salary
FROM emp
GROUP BY deptno
HAVING COUNT(*) >= 4;
```

3. WAQTD number of employees earning salary > 1200 in each job and total salary of each job must exceed 3800

```
SELECT job,
       COUNT(*) AS no_of_employees,
       SUM(sal) AS total_salary
FROM emp
WHERE sal > 1200
GROUP BY job
HAVING SUM(sal) > 3800;
```

4. WAQTD deptno and number of employees only if there are 2 MANAGERS in each dept

```
SELECT deptno, COUNT(*) AS no_of_managers
FROM emp
WHERE job = 'MANAGER'
```

GROUP BY deptno
HAVING COUNT(*) = 2;

5. WAQTD job and maximum salary of employees if the maximum salary exceeds 2600

SELECT job, MAX(sal) AS max_salary
FROM emp
GROUP BY job
HAVING MAX(sal) > 2600;

6. WAQTD the salaries which are repeated in EMP table

SELECT sal, COUNT(*) AS no_of_times
FROM emp
GROUP BY sal
HAVING COUNT(*) > 1;

7. WAQTD the hiredates which are duplicated in EMP table

SELECT hiredate, COUNT(*) AS no_of_times
FROM emp
GROUP BY hiredate
HAVING COUNT(*) > 1;

8. WAQTD average salary of each dept if average salary is less than 3000

SELECT deptno, AVG(sal) AS avg_salary
FROM emp
GROUP BY deptno
HAVING AVG(sal) < 3000;

9. WAQTD deptno if there are at least 3 employees whose name contains 'A' or 'S'

SELECT deptno, COUNT(*) AS no_of_employees


```
FROM emp
WHERE ename LIKE '%A%'
      OR ename LIKE '%S%'
GROUP BY deptno
HAVING COUNT(*) >= 3;
```

10. WAQTD min and max salaries of each job if min salary > 1000 and max salary < 5000

```
SELECT job,
       MIN(sal) AS min_salary,
       MAX(sal) AS max_salary
FROM emp
GROUP BY job
HAVING MIN(sal) > 1000
      AND MAX(sal) < 5000;
```